

Evaluation design conditions the expert-vs-auto MeSH gap: a controlled comparison of bag-of-words and BiomedBERT on the Cohen benchmark

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Abstract Systematic review screening classifiers commonly augment abstracts with Medical Subject Headings (MeSH) drawn from either expert annotation or mechanical lookup. These two sources have not been compared directly as classifier features, and no prior work has asked whether that comparison’s outcome depends on how the classifier is evaluated. Using the [Cohen et al. \(2006\)](#) drug-class benchmark on three topics, we characterise both a bag-of-words logistic regression classifier (seven reruns) and BiomedBERT (five seeds), then run robustness analyses on the Statins finding. Under the canonical 5-fold full-corpus design, the bag-of-words expert-vs-auto gap on Statins is +0.096 WSS@95%. Stratified subsampling to matched corpus size ($n = 803$) shrinks this to +0.033 (95% bootstrap CI includes zero); 10-fold cross-validation at full corpus size shrinks it to +0.021 (CI narrowly excludes zero). BiomedBERT under canonical evaluation gives +0.020, within sampling noise of the bag-of-words 10-fold result. An empirical power analysis shows that a Statins-sized effect at Opioids’ or ADHD’s per-fold variance would not have been detectable under the present per-topic fold counts ($MDE_{\text{Opioids}} = 0.189$, $MDE_{\text{ADHD}} = 0.286$), so cross-topic nulls are design-limited rather than informative. The canonical design is the condition under which the gap is most cleanly detected, and the gap attenuates under designs that increase per-fold training volume. A representation asymmetry (15.1% of Statins inputs exceed BiomedBERT’s 512-token cap in the expert-MeSH mode, and truncation preferentially removes the appended MeSH terms) may contribute to the transformer’s smaller gap but cannot be isolated from the training-volume effect without a no-truncation comparison. For screening pipelines using transformer classifiers or 10-fold bag-of-words, the observed expert-vs-auto gap on the topics tested is approximately 0.02 WSS@95%, with confidence intervals spanning zero on at least one bound.

1 Introduction

Systematic reviews assemble published evidence relevant to a clinical question. Screening reviewers read thousands of titles and abstracts to identify the small subset relevant to the review; the work is expensive and slow, and automated screening tools have been studied since at least [Cohen et al. \(2006\)](#). Their drug-class benchmark established a 15-topic testbed and an evaluation metric, work saved over sampling at 95% recall (WSS@95%), that the field continues to use. The standard input representation for screening classifiers has been an article’s title and abstract, often augmented with MeSH headings drawn from the National Library of Medicine’s controlled vocabulary. The augmentation captures topic information the abstract may not state explicitly, such as the indication that an article is about a specific drug class even when no specific drug name appears in the title.

MeSH terms are produced by two mechanisms. Professional indexers at the National Library of Medicine assign MeSH headings to each PubMed-indexed arti-

cle based on its full content; these expert annotations are the canonical reference. Automated tools separately assign MeSH terms mechanically through string matching, concept linking, or model-based prediction. Whether the two annotation sources are interchangeable as classifier features has practical importance: expert annotation is delayed by weeks or months after publication, while mechanical assignment is instantaneous and can be applied to preprints or recently-published articles that lack expert indexing. If the two produce comparable downstream classifier performance, mechanical assignment is a viable substitute. If they do not, systematic review screening depends on annotation latency in a way that limits coverage of recent literature.

To our knowledge, no prior study has isolated annotation mechanism while holding MeSH vocabulary constant. Existing work treats expert MeSH as ground truth and auto-MeSH prediction as an algorithmic problem in its own right ([Mao and Lu, 2017](#); [Jin and Szolovits, 2018](#)). Studies that augment classifier inputs with MeSH features typically use the

expert-assigned version without varying the assignment mechanism (Cohen, 2008; Matwin et al., 2010). The nearest analogue concerns features more broadly: Scott and Matwin (1999) compared phrases, synonyms, and hypernyms as features for Reuters text classification. The interaction with evaluation design has not been raised at all; the canonical 5-fold cross-validation at native topic size is treated throughout this literature as a methodological default rather than an experimental variable.

This paper takes up both questions through a controlled experiment on three Cohen topics, holding the MeSH vocabulary constant and varying only the assignment mechanism. Two classifier families serve as the test platform: a bag-of-words logistic regression classifier and BiomedBERT (formerly PubMedBERT) (Gu et al., 2021), a domain-pretrained contextual transformer. We characterise each topic-classifier combination as a distribution — seven-rerun for bag-of-words, five-seed for BiomedBERT — and supplement the main comparison with two robustness analyses on the Statins finding (subsampling to matched corpus size; 10-fold cross-validation at full corpus size) and an empirical power analysis on the per-topic variances.

Our research questions and paired hypotheses are:

- **RQ1 / H1.** Does mechanical MeSH assignment produce comparable WSS@95% to expert MeSH assignment as classifier features, holding vocabulary and classifier constant? *H1: expert MeSH produces a positive advantage under the canonical design on Statins, replicating prior work.*
- **RQ2 / H2.** Does the answer depend on the classifier? *H2: BiomedBERT reduces the expert-vs-auto gap relative to bag-of-words at Statins.*
- **RQ3 / H3.** Does the answer depend on the evaluation design? *H3: the Statins expert-vs-auto gap magnitude is conditional on per-fold training volume and per-topic corpus size.*

Under the canonical design the expected Statins advantage appears; BiomedBERT attenuates it, and the bag-of-words advantage itself shrinks under both robustness analyses to magnitudes within sampling noise of the BiomedBERT result. We do not adjudicate whether the transformer effect is architecturally distinct from the design-sensitivity effect; the data are consistent with a common training-volume mechanism affecting both. The cross-topic nulls at Opioids and ADHD are design-limited: the empirical MDE at those topics exceeds the Statins effect size, so they are not informative about the underlying mechanism.

We make three contributions.

- **A design-conditional characterisation of a widely-cited screening finding.** We show

that the Statins expert-vs-auto MeSH gap on the Cohen benchmark attenuates approximately threefold under matched-corpus-size subsampling and approximately fivefold under 10-fold cross-validation, decomposing the canonical-design magnitude into distinct corpus-size and per-fold-training-volume components.

- **A controlled cross-classifier comparison of annotation mechanism.** Holding vocabulary and evaluation constant, we compare bag-of-words and BiomedBERT on three Cohen drug-class topics, treating each topic-classifier combination as a distribution rather than a point estimate through seven-rerun and five-seed multi-run protocols with archived per-fold values.
- **A power-analytic bound on the cross-topic null.** The empirical MDE at Opioids' and ADHD's per-fold variances exceeds the Statins effect size, so the cross-topic null does not distinguish absence of a mechanism effect from absence of detection power. We report this as a design limitation rather than a substantive finding.

2 Related work

Cohen et al. (2006) introduced the drug-class benchmark of 15 systematic review topics and reported WSS@95% values for a voting perceptron classifier with binary bag-of-words features. Their cross-topic average was approximately 18.5%, with four topics showing WSS@95% below 5%. Subsequent work has continued to use this benchmark. Cohen (2008) reported SVM-based results with n-gram features and MeSH terms, using AUC rather than WSS@95%. Matwin et al. (2010) used factorised complement naive Bayes and reported WSS@95% improvements on several topics. Across this body of work, MeSH augmentation appears as a feature source but its assignment mechanism is treated as fixed, and the 5-fold cross-validation at native topic size is treated as a methodological default rather than an experimental variable.

Recent screening work has shifted toward transformer and large-language-model classifiers, with reported performance for GPT-class systems on systematic review tasks that often exceeds earlier statistical baselines (Guo et al., 2024; Alshami et al., 2023; Wang et al., 2024). This shift creates a positioning question for any new work using simpler classifiers: why study bag-of-words when the state of the art has moved on? Our answer treats bag-of-words as a controlled testbed for isolating a mechanism rather than a competing system for screening practice. The mechanism question we investigate is independent of which classifier currently performs best, because the question concerns whether

two annotation sources are interchangeable as inputs, not which input pipeline produces the highest screening performance overall.

The reliability of biomedical NLP benchmarks under label variation has received recent attention (Plank, 2022); the present paper extends this concern to evaluation-design variation. We show that a widely-cited benchmark result is stable under some perturbations of the canonical evaluation and unstable under others, and that this instability has been invisible because the field has retained the canonical design as a default.

MeSH-based feature augmentation appears in the broader text classification literature as one instance of ontology-driven feature engineering. The semasiological versus onomasiological distinction in lexical semantics (Baldinger, 1980; Geeraerts, 2010), with origins in de Saussure (1959), describes two directions of vocabulary access, and distributional semantics (Harris, 1954; Sahlgren, 2008; Lenci, 2018) provides the theoretical underpinning for bag-of-words methods. Whether that distinction maps onto the mechanical-versus-expert contrast is one interpretive frame for the canonical-design Statins finding; because the present experiments do not test it, we defer it to Section 5.2.

Biomedical transformer models (Gu et al., 2021; Lee et al., 2020; Yasunaga et al., 2022) are pretrained on PubMed abstracts and have produced state-of-the-art results on biomedical NLP benchmarks including the BLURB suite. We use BiomedBERT (rather than BioBERT, BioLinkBERT, or ClinicalBERT) because BLURB benchmarking placed BiomedBERT-base above alternatives at comparable parameter count (Gu et al., 2021) and because our comparison is not intended to identify the strongest screening architecture; we need a domain-pretrained contextual transformer with tractable fine-tuning cost. Their use for systematic review screening has been less studied than their use for biomedical question answering or named entity recognition. We use BiomedBERT as a bridge from the controlled bag-of-words experiment to representations that are closer to current screening practice.

3 Methods

3.1 Benchmark and topic selection

We use three drug-class topics from the Cohen et al. (2006) benchmark: Statins, Opioids, and ADHD. The benchmark provides PubMed identifiers and inclusion labels from completed systematic reviews. Article texts and MeSH terms were retrieved through the NCBI Entrez API and cached locally to ensure reproducibility. Table 1 summarises topic characteristics.

The three topics span different inclusion rates and

Topic	Total articles	Included	Inclusion rate
Statins	2,744	152	5.5%
Opioids	1,772	43	2.4%
ADHD	803	83	10.3%

Table 1: Cohen drug-class topics used in this study.

different clinical domains (cardiovascular, analgesic, behavioural). The Statins topic is the most studied in the screening literature and has been used as the primary reference in several follow-up papers. Opioids and ADHD differ from Statins in both corpus size and inclusion rate, providing cross-topic variation on dimensions that prior work has shown to affect classifier performance (Cohen, 2008; Matwin et al., 2010). The Cohen benchmark structure assigns each topic its native size, and the literature has retained this convention; we follow it in the main comparison and explicitly relax it in the robustness analysis (Section 3.6).

3.2 Text modes

Each article is represented in four alternative input forms:

1. `abstract`: the abstract text alone.
2. `title_abstract`: the article title concatenated with the abstract.
3. `title_abstract_mesh`: title plus abstract plus expert-assigned MeSH terms from the PubMed indexer, concatenated as space-separated tokens.
4. `auto_mesh`: abstract plus mechanically-assigned MeSH terms, where the assignment uses case-insensitive substring matching of MeSH headings against a topic-wide vocabulary built from the union of all MeSH terms observed in the topic’s cached records.

Modes 1 and 2 establish baseline performance without MeSH augmentation. Modes 3 and 4 instantiate the central manipulation: both augment the text with MeSH vocabulary, but mode 3 uses the expert assignment and mode 4 uses the mechanical assignment. The MeSH vocabulary in mode 4 is built from the topic-specific cache rather than the full MeSH thesaurus, which is the construction that makes the comparison fair. Both modes draw from the same set of available MeSH terms with different assignment mechanisms.

3.3 Classifiers

The bag-of-words pipeline tokenises with a Keras Tokenizer using stop-word removal and unigram binary

features, feeds these into a regularised logistic regression classifier, and chooses the L2 regularisation strength by cross-validation on the training folds. We report the regularised variant throughout. Full pipeline implementation appears in the project repository.

The BiomedBERT pipeline uses `microsoft/BioMedNLP-BiomedBERT-base-uncased-abstract`, the abstract-pretrained variant of BiomedBERT (formerly PubMedBERT) (Gu et al., 2021). Hyperparameters were chosen as BERT-fine-tuning defaults rather than topic-tuned: learning rate 2e-5, batch size 16, 3 fine-tuning epochs, maximum sequence length 512 tokens. Token-length analysis of the input corpus shows that 4.6–7.9% of abstracts across the three topics exceed this 512-token cap in the `abstract` and `title_abstract` modes and are truncated; the rate rises to 15.1% for Statins, 10.4% for Opioids, and 11.8% for ADHD in the `title_abstract_mesh` mode, where the MeSH terms appended at the end of the input are most affected by truncation. The bag-of-words pipeline is unbounded and sees the full input in every case. This representation asymmetry is discussed in Sections 5.1 and 6.

Class weights followed sklearn’s balanced formulation, which is necessary at the inclusion rates the Cohen topics present. Without balanced weighting, BiomedBERT produced near-random discrimination ($AUC \approx 0.43$) on the most imbalanced topic in pilot validation. Additional implementation details (mixed-precision training, seed argument mechanics, per-fold output format) are given in Appendix A.

The choice of BiomedBERT rather than BioLinkBERT, ClinicalBERT, or BiomedBERT-large is deliberate. Our aim is not to identify the strongest screening architecture but to compare annotation-mechanism sensitivity across a bag-of-words baseline and a domain-pretrained contextual transformer at tractable fine-tuning cost. BiomedBERT-base places above BioBERT on BLURB (Gu et al., 2021) at comparable parameter count, and the abstract-pretrained variant matches our input domain. Whether the finding generalises to other transformer architectures is an open question flagged in Section 6.

3.4 Evaluation

We report WSS@95%, the work saved over sampling at 95% recall (Cohen et al., 2006), as the primary metric. WSS@95% measures the fraction of the corpus a reviewer would not have to read after triaging by classifier-ranked probability, conditional on achieving 95% recall of included articles. This metric is operational rather than statistical: it reflects the workload reduction a systematic review team would achieve if they used the classifier output to prioritise screening. We also report accuracy and ROC AUC, but these are less informative for the screening application because

they treat the two error directions symmetrically while screening practice does not.

Five-fold stratified cross-validation was applied to each topic in the main comparison. Stratification preserves the inclusion rate across folds, which matters at the 2.4% inclusion rate of Opioids where a non-stratified split would risk folds with no positive examples.

3.5 Statistical analysis and reproducibility characterisation

The per-fold WSS@95% values from each topic-mode combination form the unit of analysis. For each topic and each classifier, the per-fold paired difference between expert mode and auto mode (`title_abstract_mesh` minus `auto_mesh`) gives the observation set. Bag-of-words observations under the main protocol comprise seven reruns of five folds each, producing 35 paired differences per topic and 105 pooled across the three topics. BiomedBERT observations comprise five seeds of five folds each, producing 25 paired differences per topic and 75 pooled across the three topics.

Bootstrap percentile confidence intervals were computed with 10,000 resamples of the per-fold differences with replacement. The 2.5th and 97.5th percentiles of the resampled means give the 95% CI. Paired permutation tests on the per-fold differences enumerate sign-flip patterns exactly for small samples and approximate by sampling for larger ones (Phipson and Smyth, 2010). The Nadeau and Bengio (2003) corrected resampled *t*-test addresses the variance underestimation that affects naive paired *t*-tests on *k*-fold CV data when training folds overlap.

Dependence structure and statistical unit. The per-fold paired differences are the unit of analysis. Because folds share training data — five folds share four fifths of it within a rerun, and all folds share the full corpus across reruns — these differences are positively correlated, and the bootstrap and permutation procedures are conservative under that dependence rather than exact. We report bootstrap CIs as uncertainty estimates and permutation *p*-values as complementary evidence, not as formal decision procedures, and the Nadeau-Bengio correction is included where a corrected *t*-test is the natural alternative. Multi-run and multi-seed characterisation additionally treats each topic-classifier combination as a distribution: run-level and seed-level means bound what a single-run point estimate can conceal.

Non-determinism characterisation, motivating the multi-run and multi-seed protocols. A reproducibility audit conducted earlier in this work surfaced two non-determinism findings that motivated the

protocols this paper uses. The bag-of-words pipeline produces different per-fold WSS values across identical-command reruns (drift up to 0.03 WSS@95% per fold), and two BiomedBERT analyses on Statins with different random-state initialisations gave per-fold values that drifted by up to 0.28 WSS@95%. The mechanistic sources (Keras layer initialisation under the current TensorFlow build; numpy RNG state at fine-tuning start) are documented in Appendix A. What matters here is the methodological consequence: point-estimate reporting can conceal both run-to-run and seed-to-seed variation of magnitudes that approach the effect sizes under discussion. The multi-run bag-of-words and multi-seed BiomedBERT protocols reported in Sections 4.1 and 4.2 treat each topic-classifier combination as a distribution. The same recognition prompted the design-sensitivity analysis: if concealed variation approached the effect size, so might variation across evaluation-design choices.

3.6 Robustness analyses on the Statins finding

The main comparison adopts the canonical Cohen evaluation design (5-fold stratified cross-validation at each topic’s native corpus size). The Statins canonical result — an expert-vs-auto MeSH gap that does not arise on Opioids or ADHD — raises two questions the main design cannot answer. First, is the Statins effect a function of the larger corpus that Statins has relative to Opioids (1.6×) and ADHD (3.4×), rather than a topic-intrinsic property? Second, does the 5-fold fold count interact with the effect? The 5-fold design trains each model on 80% of the topic’s articles; the alternative 10-fold design trains on 90%. We ran two robustness analyses to address each question separately.

The **matched-corpus-size analysis** stratified-subsampled Statins to $n = 803$ articles, matching the ADHD corpus size. Seven subsample seeds were drawn, each preserving the Statins inclusion rate (5.5%); each subsample was then run through the standard 5-fold pipeline with all four text modes, producing $7 \times 5 = 35$ per-fold paired differences on the subsampled Statins.

The **10-fold sensitivity analysis** retained Statins at full $n = 2,744$ but switched the cross-validation design from 5-fold to 10-fold, with seven reruns to match the existing multi-run protocol structure. This produces $7 \times 10 = 70$ per-fold paired differences. We did not run 10-fold on Opioids or ADHD because per-fold included-article counts at those topics under 10-fold (≈ 4 for Opioids, ≈ 8 for ADHD) are too low for WSS@95% to behave stably.

We also computed an **empirical power analysis** on the main-comparison per-fold variances: at the per-topic standard deviations observed under the canon-

ical 5-fold design, what is the minimum detectable effect (MDE) at 80% power for the present per-topic fold counts? This answers a question complementary to the subsampling: even if a Statins-sized effect existed at Opioids or ADHD, would the present sample size and variance allow detection?

4 Results

4.1 Bag-of-words results: topic-stratified multi-run characterisation

Tables 2, 3, and 4 report per-fold expert-vs-auto MeSH WSS@95% differences for the bag-of-words logistic regression classifier across all three topics, each characterised across seven reruns of the same pipeline configuration. The seven reruns per topic use identical arguments, code, input data, and fold-splitting seeds. The run-to-run variation reflects residual non-determinism in Keras Dense layer initialisation that the pipeline’s explicit seeding does not fully control, documented in Section 3.5.

Under the canonical evaluation design, the Statins distribution is tight and positive: per-run mean +0.096, standard deviation 0.015 across seven reruns (standard error of the mean approximately 0.006), range [+0.077, +0.114], all seven run-level gaps positive, 32 of 35 fold-level paired differences strictly positive. The two negative folds at Statins (Run 3 fold 2 at -0.053 , Run 3 fold 4 at -0.022) are small in magnitude relative to the per-fold WSS standard deviation of approximately 0.07 across Statins folds. The originally published value of +0.121 lies above the observed multi-run range upper bound of +0.114, consistent with being a single realisation from the same underlying distribution.

The Opioids and ADHD distributions are not positive and not tight. On Opioids the per-run mean is +0.007 with per-run range $[-0.019, +0.038]$, five of seven runs positive, 23 of 35 folds positive. On ADHD the per-run mean is +0.006 with per-run range $[-0.035, +0.043]$, four of seven runs positive, 18 of 35 folds positive. The per-fold gaps span both signs widely: Opioids fold gaps range from -0.411 to $+0.274$, ADHD fold gaps range from -0.310 to $+0.354$. Per-fold standard deviations within topic are 0.17 (Opioids) and 0.16 (ADHD), which is approximately 25 times the magnitudes of the run-level mean gaps. Under the canonical design, the gap is not detectable in either direction at these two topics.

4.2 BiomedBERT results: multi-seed characterisation across three topics

BiomedBERT was characterised across five seeds (42, 7, 13, 21, 31) for each of the two H-Pub2-critical modes

Run	Expert WSS mean	Auto WSS mean	Per-fold gaps	Run gap
1	0.204	0.126	[+0.089, +0.027, +0.155, +0.000, +0.120]	+0.078
2	0.198	0.109	[+0.082, +0.004, +0.155, +0.073, +0.131]	+0.089
3	0.218	0.142	[+0.089, -0.053, +0.206, -0.022, +0.164]	+0.077
4	0.200	0.098	[+0.022, +0.024, +0.191, +0.148, +0.124]	+0.102
5	0.213	0.115	[+0.080, +0.027, +0.180, +0.062, +0.144]	+0.099
6	0.216	0.105	[+0.086, +0.055, +0.158, +0.111, +0.144]	+0.111
7	0.227	0.113	[+0.054, +0.127, +0.160, +0.100, +0.131]	+0.114
Mean	0.211	0.115	—	+0.096

Table 2: BoW Statins multi-run characterisation, canonical 5-fold full-corpus design. Seven reruns of identical pipeline configuration.

Run	Expert WSS mean	Auto WSS mean	Per-fold gaps	Run gap
1	0.277	0.279	[-0.296, +0.025, -0.077, +0.274, +0.065]	-0.002
2	0.267	0.267	[-0.262, +0.023, +0.087, -0.048, +0.201]	+0.000
3	0.310	0.303	[-0.304, +0.098, -0.070, +0.156, +0.155]	+0.007
4	0.295	0.288	[-0.183, +0.031, -0.045, +0.139, +0.090]	+0.006
5	0.295	0.257	[-0.214, +0.022, +0.057, +0.200, +0.125]	+0.038
6	0.318	0.302	[-0.313, +0.073, +0.026, +0.082, +0.212]	+0.016
7	0.306	0.325	[-0.411, +0.040, -0.051, +0.184, +0.141]	-0.019
Mean	0.295	0.289	—	+0.007

Table 3: BoW Opioids multi-run characterisation, canonical 5-fold full-corpus design.

(title_abstract_mesh and auto_mesh) on each of the three topics. This produces 25 per-fold WSS@95% values per topic per mode (5 seeds $\times$ 5 folds), and 75 pooled across the three topics per mode. Tables 5, 6, and 7 report the per-seed expert and auto WSS@95% means and the per-seed expert-vs-auto gap for each topic.

Under the canonical 5-fold full-corpus evaluation, the BiomedBERT Statins per-seed gap distribution centres above zero (pooled +0.020, per-seed range [-0.007, +0.060]), but the central tendency is substantially below the bag-of-words Statins multi-run mean of +0.096. The Opioids per-seed gap distribution has a negative central tendency (-0.048 pooled, four of five seeds negative) with one seed-level outlier (seed 13 at +0.082) that we report rather than exclude. The ADHD per-seed gap distribution is centred near zero (pooled +0.003, signs split 2 positive, 2 negative, 1 near zero).

The seed 13 Opioids result deserves explicit comment. Across five seeds with identical hyperparameters, fold splits, and training data, the per-seed gap ranges from -0.127 (seed 21) to +0.082 (seed 13), a swing of 0.21 WSS@95% attributable to random initialisation alone. This seed-level swing exceeds the topic-level MDE that the power analysis in Section 4.4 reports (0.189), meaning that seed-level instability at Opioids dominates any true effect of the size the canonical Statins finding provides for reference. The four-seed mean excluding seed 13 would be -0.081; the five-seed pooled mean retaining seed 13 is -0.048. We retain seed

13 because excluding it would amount to picking the result from a five-seed distribution post-hoc, and the wider distribution itself is part of the methodological observation: at this corpus size and inclusion rate (Opioids has 43 included articles in 1,772), per-seed fluctuation is substantial enough that single-seed characterisation can mislead about both magnitude and direction.

4.3 Statistical tests

Table 8 reports the bootstrap CIs and exact paired permutation p -values for the expert-minus-auto difference, per topic and pooled, separately for each classifier family, under the canonical 5-fold full-corpus design.

The bag-of-words Statins gap is statistically detectable at the conventional $\alpha = 0.05$ threshold (CI excludes zero, permutation $p < 0.001$). The bag-of-words Opioids and ADHD gaps are not (both CIs include zero, permutation $p > 0.8$). The bag-of-words pooled across the three topics retains a small positive central tendency dominated by the Statins contribution and just excludes zero on the lower bound of the CI; the pooled permutation p -value of 0.013 reflects the Statins component rather than a consistent cross-topic effect.

The BiomedBERT per-topic gaps are not statistically detectable at any topic. The Statins CI includes zero on the lower bound, the Opioids CI on the upper bound, and the ADHD CI is centred on zero. The three-topic pooled BiomedBERT CI of [-0.039, +0.024]

Run	Expert WSS mean	Auto WSS mean	Per-fold gaps	Run gap
1	0.429	0.406	[+0.168, -0.174, +0.180, +0.081, -0.138]	+0.023
2	0.442	0.407	[+0.254, -0.249, +0.075, +0.100, -0.006]	+0.035
3	0.469	0.461	[+0.354, -0.199, -0.006, +0.013, -0.119]	+0.009
4	0.387	0.393	[+0.180, -0.255, +0.081, +0.037, -0.075]	-0.006
5	0.408	0.436	[+0.143, -0.174, -0.013, +0.050, -0.144]	-0.028
6	0.434	0.391	[+0.267, -0.162, +0.099, +0.057, -0.044]	+0.043
7	0.360	0.395	[+0.217, -0.310, -0.056, +0.062, -0.087]	-0.035
Mean	0.418	0.413	—	+0.006

Table 4: BoW ADHD multi-run characterisation, canonical 5-fold full-corpus design.

Seed	Expert WSS mean	Auto WSS mean	Gap	Seed	Expert WSS mean	Auto WSS mean	Gap
42	0.275	0.273	+0.002	42	0.601	0.656	-0.055
7	0.256	0.253	+0.004	7	0.578	0.547	+0.031
13	0.185	0.144	+0.040	13	0.606	0.544	+0.062
21	0.342	0.282	+0.060	21	0.618	0.610	+0.008
31	0.307	0.314	-0.007	31	0.639	0.667	-0.029
Pooled (25 folds)	0.273	0.253	+0.020	Pooled (25 folds)	0.608	0.605	+0.003

Table 5: BiomedBERT Statins multi-seed characterisation. Five seeds, five folds each.

Seed	Expert WSS mean	Auto WSS mean	Gap
42	0.367	0.441	-0.073
7	0.362	0.421	-0.059
13	0.438	0.357	+0.082
21	0.402	0.529	-0.127
31	0.276	0.340	-0.063
Pooled (25 folds)	0.369	0.417	-0.048

Table 6: BiomedBERT Opioids multi-seed characterisation. Five seeds, five folds each. Seed 13 produces a gap of +0.082 against the other four seeds’ range of $[-0.127, -0.059]$. The outlier is retained without exclusion.

is centred essentially on zero and does not overlap the bag-of-words Statins distribution under the canonical design.

4.4 Evaluation design sensitivity

Two robustness analyses interrogate the Statins canonical finding by varying the evaluation design while holding the classifier, the data, and the assignment mechanism constant. Figure 1 and Table 9 report the results. The matched-corpus-size analysis subsamples Statins to $n = 803$ articles, matching ADHD’s corpus size, and runs the standard 5-fold protocol across seven subsample seeds. The 10-fold sensitivity analysis retains Statins at full $n = 2,744$ but uses 10-fold instead of 5-fold cross-validation across seven reruns.

Table 7: BiomedBERT ADHD multi-seed characterisation. Five seeds, five folds each.

Three findings emerge. First, the Statins gap is approximately three times smaller at matched corpus size (+0.033 versus the canonical +0.096) and the 95% bootstrap CI includes zero. Subsampled Statins and full-size ADHD are not statistically distinguishable, in the sense that the CI of the former ($[-0.022, +0.092]$) lies well within the CI of the latter ($[-0.046, +0.060]$). Second, at full corpus size but with 10-fold cross-validation, the Statins gap shrinks to +0.021, with a CI that excludes zero only by a small margin ($[+0.001, +0.041]$). Third, the 10-fold Statins result and the canonical BiomedBERT Statins result are within sampling noise of each other: BoW 10-fold +0.021 versus BERT 5-fold +0.020, a difference of 0.001 WSS@95%. Both are an order of magnitude below the canonical BoW 5-fold reference.

The bag-of-words classifier therefore produces a substantially different expert-vs-auto gap depending on two evaluation choices that prior work treats as methodological defaults: how the corpus is sized (native versus matched) and how the cross-validation is partitioned (5-fold versus 10-fold). The canonical Cohen design happens to be the one under which the gap is most cleanly detected. We do not claim that the canonical design is the wrong default; we report that the magnitude of the Statins finding is partly a function of this default rather than an intrinsic property of the topic-classifier-mechanism combination.

We also computed an empirical power analysis on the per-fold variances at each topic under the canonical design. Table 10 reports the minimum detectable effect

Classifier	Topic	n folds	Mean diff	95% bootstrap CI	Perm p
BoW (multi-run)	Statins	35	+0.0957	[+0.075, +0.116]	< 0.001
	Opioids	35	+0.0066	[−0.050, +0.061]	0.82
	ADHD	35	+0.0059	[−0.046, +0.060]	0.83
	Pooled	105	+0.0361	[+0.009, +0.063]	0.013
BERT (multi-seed)	Statins	25	+0.020	[−0.011, +0.052]	0.19
	Opioids	25	−0.048	[−0.116, +0.020]	0.31
	ADHD	25	+0.003	[−0.040, +0.046]	0.81
	Pooled	75	−0.008	[−0.039, +0.024]	0.60

Table 8: Statistical analysis of the expert–auto WSS@95% difference by classifier and topic under the canonical 5-fold full-corpus design. Bootstrap CIs from 10,000-resample percentile bootstrap. Permutation p -values from exact sign-flip enumeration where feasible and approximate sampling at 100,000 permutations otherwise.

Design	n folds	Mean gap	95% bootstrap CI	SD
Statins, 5-fold, full $n = 2,744$ (canonical, reference)	35	+0.096	[+0.075, +0.116]	0.067
Statins, 5-fold, subsampled $n = 803$	35	+0.033	[−0.022, +0.092]	0.177
Statins, 10-fold, full $n = 2,744$	70	+0.021	[+0.001, +0.041]	0.086
Opioids, 5-fold, multi-run reference $n = 1,772$	35	+0.007	[−0.050, +0.061]	0.170
ADHD, 5-fold, multi-run reference $n = 803$	35	+0.006	[−0.046, +0.060]	0.160

Table 9: Evaluation design sensitivity on the Statins expert-vs-auto MeSH gap. The canonical Statins finding under the standard Cohen design (top row) attenuates under both subsampling to ADHD-matched size (row 2) and under 10-fold evaluation at full size (row 3). The subsampled CI includes zero. The 10-fold CI is narrowly positive but the magnitude shrinks roughly fivefold relative to the canonical reference. Opioids and ADHD reference rows are the multi-run means from Tables 3–4, matching the same $n = 35$ per-fold structure as the Statins subsampled and canonical rows for commensurability.

(MDE) at 80% power for each topic.

Topic	n folds	Observed mean	Per-fold SD	MDE (80% power)
Statins	5	+0.125	0.068	0.085
Opioids	5	−0.010	0.151	0.189
ADHD	5	−0.030	0.228	0.286

Table 10: Empirical power analysis on per-fold variances at each topic. MDE computed at $\alpha = 0.05$ two-sided, 80% power, normal approximation. With the present $n = 5$ folds the t -correction shifts MDE by less than 4%.

At the Statins per-fold variance (0.068), the canonical $n = 5$ folds give a minimum detectable effect of 0.085 at 80% power, comfortably below the observed Statins effect of +0.125 (this is why it is detected). At Opioids’ larger per-fold variance (0.151), the same $n = 5$ folds give an MDE of 0.189, which is 1.5 $\times$ the Statins effect size. At ADHD’s still larger variance (0.228), the MDE is 0.286, more than 2 $\times$ the Statins effect. A Statins-sized effect at Opioids or ADHD would not have been detected at the present sample sizes and variances. The cross-topic null at Opioids and ADHD is therefore design-limited rather than informative about the underlying mechanism: the data do not allow us to distinguish between “no effect at those topics” and “effect

present but undetectable at present sample sizes.”

4.5 Pattern across topics, classifiers, and evaluation designs

Figure 2 summarises the main-comparison expert-minus-auto WSS@95% gaps across the two classifier families and three topics. Under the canonical evaluation, the cross-classifier comparison admits a topic-stratified reading: where the bag-of-words classifier shows a robust expert-vs-auto MeSH gap (Statins, +0.096 with tight distribution), BiomedBERT attenuates that gap to +0.020. Where the bag-of-words classifier shows no detectable gap (Opioids and ADHD), BiomedBERT shows the same null pattern.

The robustness analyses in Section 4.4 require this reading to be qualified in two ways. First, the bag-of-words Statins gap is itself sensitive to two evaluation parameters; under both subsampling and 10-fold the gap attenuates to magnitudes comparable to the BiomedBERT canonical result. The transformer’s apparent attenuation of the Statins gap, measured against the canonical BoW reference, becomes less distinctive when measured against the BoW result under designs that increase per-fold training volume. Second, the cross-topic null at Opioids and ADHD does not estab-

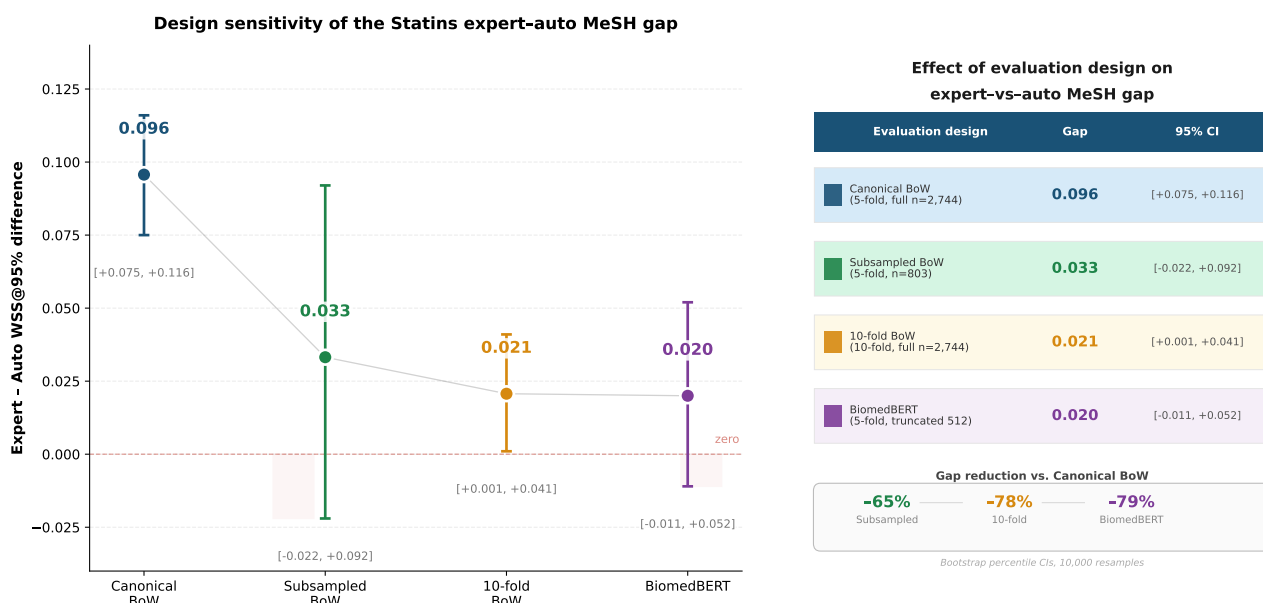


Figure 1: Design sensitivity of the Statins expert–auto MeSH gap. *Left:* Mean paired WSS@95% differences with 95% bootstrap percentile CIs (10,000 resamples). The canonical BoW result (+0.096) attenuates under matched-corpus subsampling (+0.033, CI includes zero) and 10-fold cross-validation (+0.021, CI narrowly excludes zero). BiomedBERT under canonical evaluation (+0.020, CI includes zero on lower bound) is within sampling noise of the BoW 10-fold result. Red shading indicates CIs spanning zero. *Right:* Summary table with gap magnitudes and percentage reductions relative to the canonical reference.

lish absence of a mechanism effect at those topics; the empirical MDE at their variances exceeds the Statins effect size, so the present design lacks power to detect a Statins-sized effect there if it existed.

The cleanest summary the data support is: the expert-vs-auto MeSH gap is real on Statins under the canonical Cohen evaluation design; under designs that increase per-fold training volume the gap shrinks to magnitudes comparable to those observed for BiomedBERT under the canonical design; and the cross-topic pattern at Opioids and ADHD is consistent with both “no effect at those topics” and “effect present but undetectable at present sample sizes.”

5 Discussion

5.1 What this paper shows, design-conditional reading

Under the canonical Cohen evaluation, expert MeSH outperforms mechanical MeSH by +0.096 WSS@95% across seven reruns of the same pipeline configuration, with all seven run-level means positive and 32 of 35 fold-level paired differences positive. This is a real effect at an evaluation design that is widely used in the field. The robustness analyses do not contradict this finding.

They show that its magnitude depends on at least

two evaluation design parameters the literature treats as defaults. Subsampling Statins to ADHD’s corpus size shrinks the gap roughly threefold and the CI includes zero. Switching from 5-fold to 10-fold at full corpus size shrinks the gap roughly fivefold to +0.021. The 10-fold BoW Statins result is within sampling noise of the BiomedBERT 5-fold Statins result (+0.020, difference 0.001). These observations are mutually consistent with two not-necessarily-exclusive readings.

The first reading is that the expert-vs-auto MeSH gap is a real and structural property of the topic, but that the classifier’s response to the difference depends on how much training data the classifier has effectively absorbed. At smaller per-fold training volumes, the classifier relies on the MeSH context to a degree that makes the assignment-mechanism difference observable. At larger per-fold training volumes (either through 10-fold partitioning or through the representational capacity of a transformer), the classifier extracts more signal from the abstract alone, making the MeSH context less critical and shrinking the observed gap. Under this reading, the 10-fold BoW result and the 5-fold BERT result may be explained by the same mechanism: more effective access to abstract text reduces the marginal contribution of MeSH augmentation.

The alternative is that the canonical 5-fold full-corpus result is the point estimate that happens to be most cleanly recovered from the Cohen benchmark

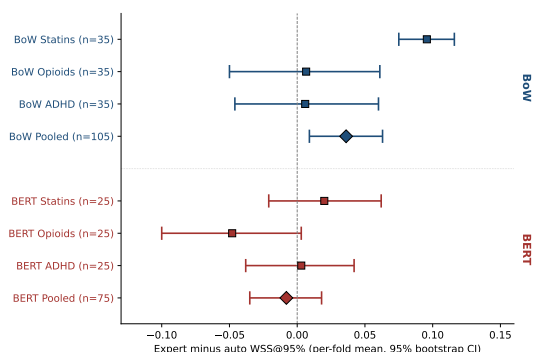


Figure 2: Expert-minus-auto MeSH WSS@95% differences across classifiers and topics, canonical design. Each row shows the per-fold mean difference (point) with a 95% bootstrap percentile confidence interval (horizontal bar). BoW rows summarise seven-rerun multi-run characterisation (35 per-fold differences per topic, 105 pooled). BERT rows summarise five-seed multi-seed characterisation (25 per-fold differences per topic, 75 pooled). The non-overlap of the BoW Statins interval [+0.075, +0.116] and the BERT Statins interval [-0.011, +0.052] is the cross-classifier finding at the topic where the gap robustly exists under canonical design. Both classifiers show confidence intervals spanning zero at Opioids and ADHD. Figure 1 shows the design-sensitivity decomposition of the BoW Statins canonical finding.

for Statins, and that the underlying effect is closer to the 10-fold magnitude. Under this reading, the historical Statins effect that the literature has reported (Cohen, 2008; Matwin et al., 2010) is an overestimate of the effect’s typical magnitude under other evaluation designs, and the present 10-fold result is an alternative point estimate the field could adopt. The CIs of the canonical and 10-fold designs do not overlap ([+0.075, +0.116] versus [+0.001, +0.041]), so the data support a real difference between the two designs rather than sampling variation alone.

We do not adjudicate between these readings. Both are consistent with the data, and either would constitute a methodologically useful warning to the screening literature that the Cohen Statins effect is more design-conditional than has been recognised. Future work that varies per-fold training volume systematically (e.g. via learning-curve analysis at varying training-set fractions) could resolve the question.

A third contributing factor to the BoW-BERT difference at Statins, distinct from training volume, is BERT’s 512-token truncation. In the `title_abstract_mesh` mode where the canonical bag-of-words gap arises, 15.1% of Statins inputs are truncated at 512 BiomedBERT subword tokens (Section 3.3), and the truncated content is preferentially the MeSH terms appended

at the end of the input. The bag-of-words classifier sees the full MeSH augmentation in every case. The direction of the resulting bias is important. Expert-assigned MeSH terms are typically more numerous than the mechanically-assigned terms produced by our substring-matching procedure, which is why the truncation rate in the expert-MeSH mode (15.1%) is elevated relative to the abstract-only base rate (4.6–7.9%). When truncation fires, expert mode loses more MeSH signal per truncated example than auto mode, precisely because expert mode had more MeSH signal to lose. Truncation therefore biases the observed BiomedBERT gap toward zero: the +0.020 result is likely conservative with respect to the true expert advantage in a no-truncation regime. This bias contribution cannot be isolated from the training-volume contribution without a no-truncation BERT comparison (e.g. Longformer, or token-budget-restructured input ordering that places MeSH terms before the abstract body), which we leave as future work and disclose as a limitation in Section 6.

5.2 One theoretical interpretation, not a demonstrated mechanism

An earlier draft of this paper offered a lexical-semantic interpretation of the canonical Statins finding, drawing on the semasiological versus onomasiological distinction (Baldinger, 1980; Geeraerts, 2010). Mechanical MeSH lookup operates semasiologically: it begins with linguistic forms in the text and finds related forms in the controlled vocabulary. Expert MeSH annotation operates onomasiologically: the indexer begins with the article’s topical content and selects vocabulary that expresses it.

We retain this as one interpretive frame, not as a demonstrated mechanism. The distinction characterises the assignment process, not the classifier’s response, and its observability in classifier outputs is conditional on the classifier-evaluation combination: where the classifier has limited effective access to the abstract, the onomasiologically-curated annotation carries more of the signal; where access is more effective, the abstract itself supplies what the augmentation would otherwise have provided. Direct evidence for this reading would require probing analysis of the transformer representations. The present experiments cannot distinguish the lexical-semantic interpretation from a purely training-volume interpretation, so we flag this as an open interpretive question rather than a claim the paper supports.

5.3 Practical implication for screening pipelines

The transformer’s +0.020 Statins gap is in the same range as the BoW 10-fold Statins gap, and both are well below the canonical BoW 5-fold reference. Under the conditions tested, a screening pipeline using BiomedBERT with mechanical MeSH would see WSS@95% performance within approximately 0.02 of the expert-MeSH version on a Statins-like topic; this is a small operational cost relative to the latency and coverage benefits of mechanical assignment. The same pipeline using bag-of-words logistic regression at 10-fold evaluation would show approximately the same cost. The classifier choice and the evaluation design therefore interact, and the literature’s reported expert-MeSH advantage of ~ 0.10 WSS@95% under the canonical design is closer to a maximum than to a typical operational value.

This implication is design-conditional. Under the canonical Cohen evaluation, the gap that the literature reports is what the pipeline sees. When the operational evaluation increases per-fold training volume, through 10-fold or through architectures that make more effective use of the abstract, the gap the pipeline sees is smaller. At topics where the expert-vs-auto gap does not robustly exist under any evaluation, as at Opioids and ADHD in our data, the substitution question is moot. Expert annotation supports retrieval, browsing, and downstream uses for which mechanical assignment may not be adequate; the present implication concerns one specific downstream use (screening classifier training) and does not generalise beyond it without further evidence.

5.4 What this finding does not establish

The data do not establish that mechanical assignment outperforms expert assignment at the BiomedBERT scale on any of the three topics. They establish that expert assignment does not produce the magnitude of advantage predicted by the bag-of-words canonical-design Statins finding when the classifier is BiomedBERT or when the bag-of-words evaluation uses a design that increases per-fold training volume. Statements such as “BERT reverses the gap” or “auto-MeSH outperforms expert MeSH” would overstate what the analysis supports.

The cross-topic null at Opioids and ADHD does not establish that no expert-vs-auto effect exists at those topics. The empirical MDE at those topics’ per-fold variances (0.189 for Opioids, 0.286 for ADHD) exceeds the canonical Statins effect size (+0.125), meaning that a Statins-sized effect at either topic would have been undetectable at the present sample size. Conclusions about whether the annotation-mechanism effect generalises across drug-class topics will require either larger

samples or smaller per-fold variances than the Cohen benchmark provides at those topics.

5.5 What the finding leaves open for representation analysis and design isolation

The present experiment shows that BiomedBERT under the canonical evaluation is approximately indifferent to the assignment mechanism on the topic where bag-of-words under the same evaluation shows the effect, and that bag-of-words under designs that increase per-fold training volume also shows the same approximate indifference. Whether this indifference reflects implicit recovery of the expert-curated structure inside the representation, an algorithmically alternative solution that achieves comparable performance without recovering that structure, or a saturation effect of training volume, cannot be determined from classification accuracy alone. Layer-wise probing of the hidden states, comparing recoverability of expert and auto MeSH structure across transformer layers and across designs where the assignment-mechanism effect varies in magnitude, is the natural next step. [Plank \(2022\)](#) frames the broader concern about benchmark evaluation under label variation that this question connects to.

Two further design-isolation experiments would clarify the picture. A learning-curve analysis at varying training-set fractions of Statins (e.g. 25%, 50%, 75%, 100% with 5-fold) would isolate the per-fold training volume effect cleanly. A matched-evaluation-design comparison running BiomedBERT at both 5-fold and 10-fold on Statins would test whether the transformer effect is itself further attenuated by additional training data, or whether it has already saturated at 5-fold. Either is a natural extension of the present work.

6 Limitations

The experimental design has limitations that bound the interpretation of the result.

The transformer comparison uses a single architecture, BiomedBERT-base. Whether the finding generalises to BioBERT, BiomedBERT-large, BioLinkBERT, or to more recent biomedical large language models is an open question. The pattern’s robustness across architectures is the natural next test.

The BiomedBERT 512-token sequence cap truncates a topic-dependent share of inputs: in the `title_abstract_mesh` mode, 15.1% of Statins, 10.4% of Opioids, and 11.8% of ADHD inputs exceed the cap, and truncation preferentially removes the MeSH terms appended at the end of the input. The bag-of-words

pipeline is unbounded and sees the full input in every case. This representation asymmetry could contribute to the smaller BiomedBERT Statins gap relative to the canonical bag-of-words reference; we cannot isolate this contribution from the training-volume contribution without a no-truncation BERT comparison (e.g. Longformer architecture or token-budget-restructured input ordering that places MeSH terms before the abstract body). We disclose this asymmetry in Section 3.3 rather than treat it as a confound the present analysis can correct for. The single-ontology constraint is similar: we tested MeSH only, and whether the same finding holds for other biomedical ontologies (SNOMED-CT, UMLS, the NEO ontology used in the original thesis work on dementia) is not established by this experiment. The English-only scope is a third constraint that follows from the Cohen benchmark itself.

Three of fifteen available Cohen topics provide the cross-topic evidence. This is reasonable coverage but not exhaustive. Extending to the remaining twelve topics would clarify the topic-dependence finding.

The evaluation design sensitivity revealed by the robustness analyses bounds the generality of the canonical-design result. We have decomposed the Statins canonical finding into a corpus-size component (the matched- n subsampling reduces the gap from +0.096 to +0.033) and a fold-count component (the 10-fold result at full corpus size reduces it to +0.021). We have not run the joint design (10-fold at matched n), which would complete the decomposition, because the resulting per-fold included-article count at $n = 803$ under 10-fold (≈ 4) is too low for WSS@95% to behave stably. The decomposition should be treated as approximate rather than exhaustive.

The BiomedBERT hyperparameters were chosen as standard fine-tuning defaults rather than optimised for the Cohen topics. Light tuning of the learning rate or number of epochs may produce modest gains in absolute WSS@95% but is unlikely to change the qualitative comparison between expert and auto modes since both use identical training procedures. We did not run BiomedBERT under 10-fold evaluation; whether the transformer effect attenuates further at 10-fold or has already saturated at 5-fold is an open question.

The bag-of-words pipeline exhibits residual non-determinism between identical-command reruns (per-fold drift up to 0.03 WSS@95%); the mechanism and mitigation approach are documented in Appendix A. Whether seven reruns are sufficient to characterise the run-to-run distribution is itself an assumption: at the observed run-to-run standard deviation of 0.015 WSS@95% on Statins, seven reruns give a standard error of the mean of approximately 0.006, small relative to the effect sizes under discussion.

The mechanical MeSH assignment procedure uses

case-insensitive substring matching against a topic-wide MeSH vocabulary built from the cache of PubMed records for that topic. Alternative procedures (MetaMap, ScispaCy, cTAKES, MedCAT) perform contextual disambiguation that substring matching does not. Whether the finding extends to these more sophisticated mechanical procedures is an additional open question.

The seed 13 result on Opioids deserves a methodological caveat. Four seeds produce gaps in $[-0.127, -0.059]$ and one (seed 13) produces +0.082. Whether this represents a single tail draw from a wide distribution or evidence of a bimodal distribution at Opioids cannot be resolved with five seeds. A future seed pool of 15 or more seeds at Opioids would resolve this. Retaining the outlier in the reported pooled mean (-0.048) is the conservative choice.

7 Conclusion

We tested whether the expert-vs-auto MeSH gap observed in bag-of-words systematic review screening on the Cohen benchmark is stable across evaluation-design perturbations and across classifier families. Under the canonical Cohen evaluation, the Statins result reproduces the effect reported by prior work on this benchmark; Opioids and ADHD show null gaps that an empirical power analysis reveals to be design-limited rather than informative. Two robustness analyses on Statins show that the canonical bag-of-words finding is conditional on the evaluation design: subsampling to matched corpus size shrinks the gap threefold with a CI that includes zero, and 10-fold cross-validation at full corpus size shrinks it approximately fivefold to a magnitude within sampling noise of the canonical BiomedBERT result.

The expert-vs-auto MeSH gap is therefore real at the canonical Cohen evaluation but evaluation-design-conditional in a way that the screening literature has not previously characterised. Where the gap robustly exists under canonical evaluation, transformer classifiers and bag-of-words classifiers under designs that increase per-fold training volume produce approximately the same attenuated gap; where the gap does not robustly exist under any tested design, the substitution question is moot. Whether the attenuation pattern generalises across transformer architectures, ontologies, and the remaining Cohen topics is an open question that Section 6 enumerates in full. The broader contribution is methodological: benchmark conclusions about feature-source effects can change substantially under reasonable evaluation-design changes, even when the underlying task and data are held constant.

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A Reproducibility and implementation details

A.1 Repository and per-fold outputs

The Cohen benchmark loader, BoW pipeline, BiomedBERT pipeline, evaluation metrics, the statistical analysis scripts (`bootstrap_paired_permutation.py`, `demo_statistical_analysis.py`, `parse_bow_multirun.py`), and the robustness-analysis scripts in

paper_experiments/ (patch_cohen_pipeline.py, run_statins_subsampling.sh, run_statins_10fold.sh, parse_bow_experiments.py, power_analysis.py) are available in the project repository. Per-fold WSS@95%, AUC, and accuracy values for all BiomedBERT runs are saved as .json sibling files (bert_{topic}_{mode}_seed{N}.json) and can be parsed deterministically. Multi-seed per-topic summaries are saved as bert_{topic}_multiseed_summary.json. The cross-topic pooled multi-seed summary is in bert_three_topic_multiseed_summary.json. Per-fold WSS@95% values for the seven BoW reruns per topic are saved in bow_{topic}_multirun_summary.json. Per-fold values for the subsampling and 10-fold robustness analyses are in paper_experiments/outputs/bow_experiments_summary.csv.

A.2 BiomedBERT implementation

Mixed-precision training (fp16) was used on T4 GPUs to make the full four-mode comparison feasible within Google Colab’s free-tier budget. The stratified fold splits and the BertConfig.seed are seeded explicitly through the --seed pipeline argument. Per-fold WSS values are written to .json sibling files alongside the standard .txt outputs for downstream parsing.

A.3 Non-determinism sources

The BoW pipeline uses nominally explicit seeds (set_seeds() in src/cohen_pipeline.py) which call np.random.seed() and tf.random.set_seed(), but does not call tf.keras.utils.set_random_seed() or tf.config.experimental.enable_op_determinism(). The consequence is residual non-determinism in Keras Dense layer initialisation. The hypothesis that observed run-to-run drift is caused by oneDNN runtime optimisations was tested by running with TF_ENABLE_ONEDNN_OPTS=0 and falsified: two consecutive identical-command runs with oneDNN disabled still produced different per-fold values, isolating the source to Keras layer initialisation under the current TensorFlow build. Rather than pursue this further into the TensorFlow stack, we characterise the run-to-run distribution as reported in Section 4.1. The BiomedBERT pipeline uses explicit --seed arguments at each invocation and the multi-seed protocol disclosed in Section 4.2; residual seed-to-seed variation of up to 0.28 WSS@95% observed in early Statins runs motivated the five-seed characterisation across all three topics.

B Hypothesis register state

The hypothesis tested in this paper, registered prior to the BiomedBERT experiment, is:

H-Pub2: The bag-of-words finding (mechanical lookup fails, expert assignment succeeds) persists when the classifier is BiomedBERT. Specifically: BiomedBERT with auto-MeSH features will not recover the WSS@95% gap relative to BiomedBERT with expert-MeSH features.

Falsifying evidence specified in advance: BiomedBERT achieves comparable WSS@95% with auto-MeSH features as with expert-MeSH features, with the gap closing to within standard deviations on at least Statins.

The disposition is partial falsification, design-conditional. At Statins under the canonical evaluation, where the bag-of-words gap robustly exists (multi-run mean +0.096, all seven reruns positive), BiomedBERT attenuates it to a multi-seed pooled +0.020. The reduction is substantial but does not close the gap to within seed-level standard deviations on the BERT side. The robustness analyses (Section 4.4) show that the bag-of-words canonical-design Statins gap itself attenuates substantially under matched-corpus-size and under 10-fold evaluation, with the 10-fold result (+0.021) within sampling noise of the canonical BERT result. The H-Pub2 framing of “transformer absorbs a structural bag-of-words effect” is therefore best read as “transformer under canonical evaluation produces a result comparable to bag-of-words under designs that increase per-fold training volume”; both attenuate the canonical-design gap to a small positive magnitude that may or may not be different from zero depending on which interval one privileges. At Opioids and ADHD, where the bag-of-words gap does not robustly exist under any tested design and an empirical power analysis shows the design lacks power to detect a Statins-sized effect at those topics’ variances, H-Pub2 was not directly testable against the assumed-positive bag-of-words baseline. The framing supported by the evidence is therefore design-conditional attenuation at Statins, with the open question of whether the attenuation is a property of the transformer specifically or of the increased effective training volume that distinguishes BERT and BoW-10-fold from the BoW canonical baseline.